

SHEN ZHENG

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4804 Forbes Ave, Pittsburgh, PA 15213

EDUCATION

Carnegie Mellon University, U.S.

Ph.D. in Robotics, Robotics Institute

2024.8 – present

Advisor: Prof. Srinivasa Narasimhan

Carnegie Mellon University, U.S.

M.S. in Computer Vision, Robotics Institute

2022.8 – 2023.12

Advisor: Prof. Srinivasa Narasimhan

Wenzhou-Kean University, China

B.S. in Mathematical Sciences

2017.9 – 2021.6

Advisor: Prof. Gaurav Gupta

RESEARCH INTEREST

Computer Vision, Generative and World Models, Efficiency, Autonomous Driving, Embodied AI

SELECTED PUBLICATIONS

MapLightning: Online Vectorized HD Map Construction with 1D Map Tokens

Shen Zheng, Anurag Ghosh, Mani Ramanagopal, Srinivasa Narasimhan

Under Review [\[arXiv\]](#)

WarpI2I: Image Warping for Image-to-Image Translation

Shen Zheng, Anurag Ghosh, Gaurav Parmar, Srinivasa Narasimhan

ECCV 2026 [\[arXiv\]](#) [\[Page\]](#)

ROADWork: A Dataset and Benchmark for Learning to Recognize, Observe, Analyze and Drive Through Work Zones

Anurag Ghosh, Shen Zheng, Robert Tamburo, Juan R. Alvarez Padilla, Hailiang Zhu, Michael Cardei, Nicholas Dunn, Christoph Mertz, Srinivasa Narasimhan

ICCV 2025 [\[arXiv\]](#) [\[Page\]](#)

Instance-Warp: Saliency Guided Image Warping for Unsupervised Domain Adaptation

Shen Zheng*, Anurag Ghosh*, Srinivasa Narasimhan

WACV 2025 [\[arXiv\]](#) [\[Page\]](#)

TPSeNCE: Towards Artifact-Free Realistic Rain Generation for Deraining and Object Detection in Rain

Shen Zheng, Changjie Lu, Srinivasa Narasimhan

WACV 2024 [\[arXiv\]](#) [\[Page\]](#)

Low-Light Image Enhancement: A Comprehensive Survey and Beyond

Shen Zheng, Yiling Ma, Jinqian Pan, Changjie Lu, Gaurav Gupta

Survey [\[arXiv\]](#)

PointNorm: Dual Normalization is All You Need for Point Cloud Analysis

Shen Zheng, Jinqian Pan, Changjie Lu, Gaurav Gupta

IJCNN 2023 Oral [\[arXiv\]](#) [\[Page\]](#)

Semantic-Guided Zero-Shot Learning for Low-Light Image/Video Enhancement

Shen Zheng, Gaurav Gupta

WACV 2022 [\[arXiv\]](#) [\[Page\]](#)

AS-IntroVAE: Adversarial Similarity Distance Makes Robust IntroVAE

Changjie Lu, Shen Zheng, Zirui Wang, Omar Dib, Gaurav Gupta

ACML 2022 [\[arXiv\]](#)

INDUSTRY EXPERIENCE

Waymo

Perception Software Engineer Intern, 3D Occupancy and World Model

Summer 2026

Mentor: Zhenyao Zhu

Waymo

Perception Software Engineer Intern, Online HD Map Construction

Summer 2025

Mentor: Guohao Zhang

Lucid Motors

Perception Software Engineer, Traffic Light Detection

2024.2 – 2024.7

Director: Dr. Feng Guo

Momenta

Perception Engineer Intern, Traffic Light Detection

2021.9 – 2022.2

Director: Dr. Wangjiang Zhu